Lesson 7.3 - Probability of a Uniform Distribution

> **Video:** Continuous Probability Uniform Distribution Problems by The Organic Chemistry Tutor **Chapter:** 7 - Continuous Distributions

Formulas (uniform on $[a, b]$)

The graph is a **rectangle** with total area 1:

$$\text{area} = \text{base} \times \text{height} = (b-a)\,f(x) = 1 \;\Rightarrow\; f(x) = \frac{1}{b-a}, \quad a \le x \le b$$

Quantity	Formula
Probability of an interval	**base × height** $= \dfrac{\text{interval width}}{b-a}$
Mean = median	$\dfrac{a+b}{2}$
Variance	$\dfrac{(b-a)^2}{12}$
Standard deviation	$\dfrac{b-a}{\sqrt{12}}$
k-th percentile value K	$\dfrac{k}{100} = \dfrac{K-a}{b-a} \;\Rightarrow\; K = a + \dfrac{k}{100}(b-a)$

Special cases:

- $P(X = c) = 0$ for any single value (a line has no width).
- $P(X > b) = P(X < a) = 0$ (the height is zero outside $[a, b]$).

Example 1 - waiting for a train (0 to 40 minutes)

The wait time is uniformly distributed on $[0, 40]$.

(a) PDF

$$f(x) = \frac{1}{b-a} = \frac{1}{40-0} = \frac{1}{40}, \qquad 0 \le x \le 40$$

(b) Graph: a flat line at height $\frac{1}{40}$ from $x = 0$ to $x = 40$, which forms a rectangle.

(c) Wait less than 8 minutes

$$P(X < 8) = \text{width} \times \text{height} = (8-0) \times \frac{1}{40} = \frac{8}{40} = \frac{1}{5} = 0.20 = 20\%$$

(d) Wait more than 30 minutes

$$P(X > 30) = \text{width} \times \text{height} = (40-30) \times \frac{1}{40} = \frac{10}{40} = 0.25 = 25\%$$

(e) Assorted probabilities

- Wait between 10 and 26 minutes:

$$P(10 < X < 26) = \frac{26-10}{40} = \frac{16}{40} = \frac{2}{5} = 0.40 = 40\%$$

- $P(X = 20) = 0$, a single value.
- $P(X > 45) = 0$, outside the range.

(f) Mean and standard deviation

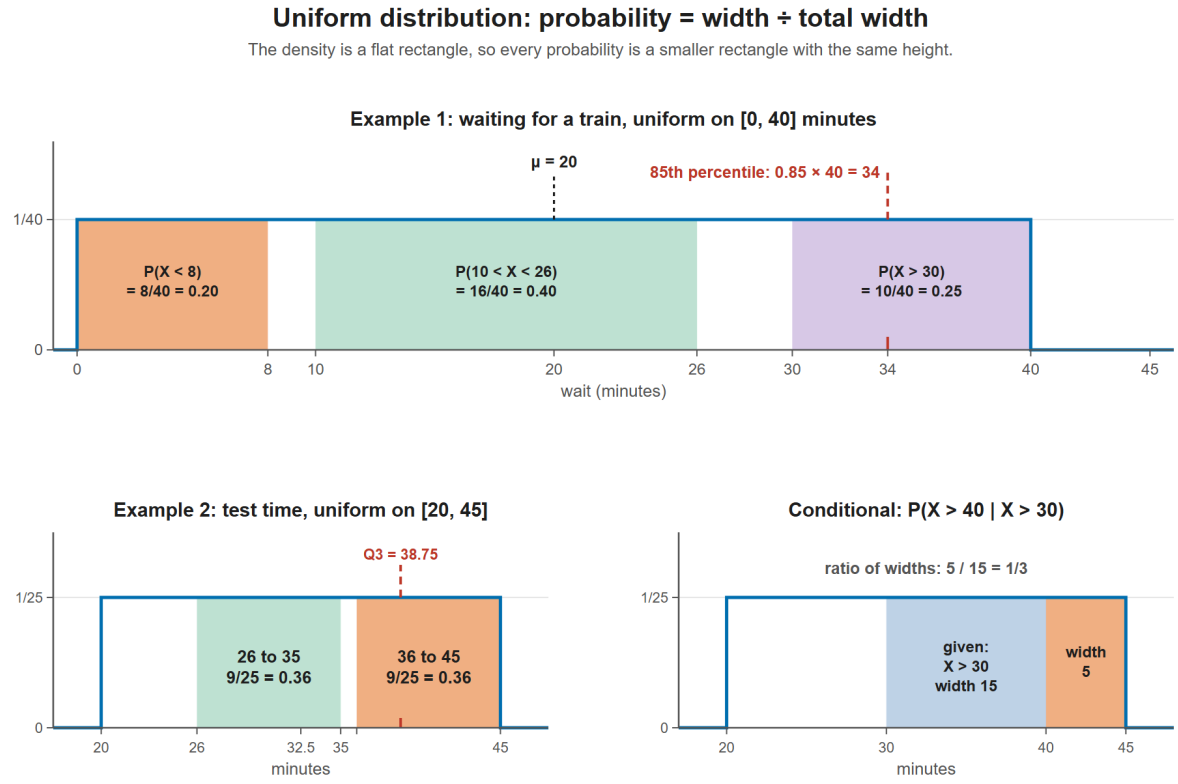
$$\mu = \frac{a+b}{2} = \frac{0+40}{2} = 20, \qquad \sigma = \frac{b-a}{\sqrt{12}} = \frac{40}{\sqrt{12}} \approx 11.55$$

(g) 85th percentile

The area to the left of K must be 0.85:

$$0.85 = (K-0)\cdot\frac{1}{40} \;\Rightarrow\; K = 0.85 \times 40 = 34 \text{ minutes}$$

This is reasonable: the 75th percentile is 30, so the 85th must lie between 30 and 40.



Each probability is the width of the shaded strip divided by the total width. Strips of equal width (26 to 35 and 36 to 45) have equal probability. For the conditional question, the given part (X > 30) becomes the new total width.

Example 2 - time to finish a chemistry test (20 to 45 minutes)

(a) PDF

$$f(x) = \frac{1}{b-a} = \frac{1}{45-20} = \frac{1}{25}, \qquad 20 \le x \le 45$$

(b) Graph: a rectangle of height $\frac{1}{25}$ from 20 to 45.

(c) More than 36 minutes

$$P(X > 36) = \frac{b - 36}{b - a} = \frac{45 - 36}{25} = \frac{9}{25} = 0.36 = 36\%$$

(d) Between 26 and 35 minutes

$$P(26 < X < 35) = \frac{\text{width}}{b - a} = \frac{35 - 26}{25} = \frac{9}{25} = 0.36$$

The width is the same as in (c), so the probability is the same.

(e) Mean (= median), variance, standard deviation

$$\mu = \frac{a + b}{2} = \frac{20 + 45}{2} = 32.5, \qquad \sigma^2 = \frac{(b - a)^2}{12} = \frac{25^2}{12} = \frac{625}{12} \approx 52.08, \qquad \sigma = \sqrt{\sigma^2} \approx 7.217$$

(f) Third quartile (Q_3 = 75th percentile)

$$0.75 = (K - 20) \cdot \frac{1}{25} \;\Rightarrow\; K - 20 = 18.75 \;\Rightarrow\; K = 38.75 \text{ minutes}$$

Check: halfway between the median 32.5 and the maximum 45 is $(32.5 + 45)/2 = 38.75$ ✓

(g) Conditional: more than 40 minutes, given more than 30 minutes

$$P(X > 40 \mid X > 30) = \frac{P(X > 40 \text{ and } X > 30)}{P(X > 30)} = \frac{P(X > 40)}{P(X > 30)}$$

If $X > 40$, then $X > 30$ is automatically true, so the intersection is just $X > 40$.

$$= \frac{(45 - 40) \cdot \frac{1}{25}}{(45 - 30) \cdot \frac{1}{25}} = \frac{5}{15} = \frac{1}{3} \approx 33.3\%$$

The heights cancel, so a conditional uniform probability is just a **ratio of widths**.

Key takeaways

- Uniform probability = **width of the interval ÷ total width**.
- A single point, or anything outside $[a, b]$, has probability 0.
- Percentile: solve $\frac{k}{100} = \frac{K - a}{b - a}$ for K.
- Mean = median = $(a + b)/2$, and $\sigma = (b - a)/\sqrt{12}$.